10. The transverse nature of light is shown in,
    (a) interference
    (b) diffraction
    (c) scattering
    (d) polarisation

Answers

1) d	**2)** b	**3)** a	**4)** c	**5)** b
6) b	**7)** a	**8)** b	**9)** c	**10)** d

Short Answer Questions

1. What are the salient features of corpuscular theory of light?
2. What are the important points of wave theory of light?
3. What is the significance of electromagnetic wave theory of light?
4. Write a short note on quantum theory of light.
5. Define wavefront.
6. What are the shapes of wavefront for (a) source at infinite, (b) point source and (c) line source?
7. State Huygens' principle.
8. What is interference of light?
9. What is phase of a wave?
10. Obtain the relation between phase difference and path difference.
11. What are coherent sources?
12. How does wavefront division provide coherent sources?
13. What is intensity (or) amplitude division?
14. How do source and images behave as coherent sources?
15. What is bandwidth of interference pattern?
16. What is diffraction?
17. Differentiate between Fresnel and Fraunhofer diffraction.
18. Discuss the special cases on first minimum in Fraunhofer diffraction.
19. What is Fresnel's distance? Obtain the equation for Fresnel's distance.
20. Mention the differences between interference and diffraction.
21. What is a diffraction grating?
22. What is resolution?
23. What is Rayleigh's criterion?
24. What is the difference between resolution and magnification?
25. What is polarisation?
26. Differentiate between polarised and unpolarised light
27. Discuss polarisation by selective absorption.
28. What are polariser and analyser?
29. What are plane polarised, unpolarized and partially polarised light?
30. State and obtain Malus' law.
31. List the uses of polaroids.
32. State Brewster's law.
33. What is angle of polarisation and obtain the equation for angle of polarisation.
34. Discuss about pile of plates.
35. What is double refraction?
36. Mention the types of optically active crystals with example.
37. Discuss about Nicol prism.
38. How is polarisation of light obtained by scattering of light?

UNIT-7(XII-Physics_Vol-2)_new.indd 103 21-07-2020 16:36:45